

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** 66 g of CO_2 occupies 22.4 L at STP.

Reason (R): 44 g of CO_2 contains 6×10^{23} molecules.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** Temperature in Kelvin scale is always positive.

Reason (R): The least possible temperature in Kelvin scale is 0 K (or) -273°C .

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Arrange the following compounds in the increasing order of number of oxygen atoms.

- | | |
|-----------------------|------------------------|
| (A) Plumbic carbonate | (B) Ferric chromate |
| (C) Zinc silicate | (D) Aluminium sulphite |
| (a) ACBD | (b) CADB |
| (c) ACDB | (d) CABD |

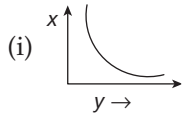
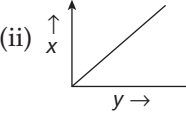
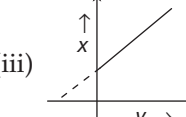
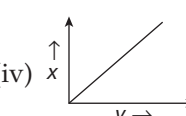
4. On heating, ammonia gives nitrogen and hydrogen. Hydrogen on treatment with chlorine gas forms hydrogen chloride gas. Hydrogen chloride gas on dissolution in water forms hydrochloric acid. Hydrochloric acid, on treatment with sodium hydroxide, forms sodium chloride and water. Sodium chloride, thus, formed on treatment with silver nitrate forms silver chloride and sodium nitrate

Various types of reactions involved in the above process are given below. Arrange them in a proper sequence as given above.

- | | |
|-------------------|--------------------|
| (A) Combination | (B) Neutralization |
| (C) Precipitation | (D) Decomposition |
| (a) ADCB | (b) ADBC |
| (c) DACB | (d) DABC |

5.8 Chapter 5 Mole Concept, Stoichiometry and Behaviour of Gases

5. Match the entries of Column A with those of Column B.

Column A	Column B
(i) 	(A) $x \rightarrow \text{volume}, y \rightarrow \frac{1}{\text{pressure}}$
(ii) 	(B) $x \rightarrow \text{volume}, y \rightarrow \text{temp}(K)$
(iii) 	(C) $x \rightarrow \text{volume}, y \rightarrow \text{pressure}$
(iv) 	(D) $x \rightarrow \text{volume}, y \rightarrow \text{temp}(^{\circ}C)$

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)
 (b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)
 (c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
 (d) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (C)

6. An oxyacid contains two hydrogen atoms, one sulphur atom and four oxygen atoms. Write the formula of the compound formed by the negative radical of the acid with ferric ion.

- (a) $\text{Fe}_2(\text{SO}_4)_3$ (b) FeSO_4
 (c) $\text{Fe}_3(\text{SO}_4)_2$ (d) $\text{Fe}(\text{SO}_4)_3$

7. If the volume occupied by a certain mass of gas is decreased to 70%, what would be the percentage change in pressure at the same temperature?

- (a) +57% (b) +70% (c) +35% (d) +43%

8. Which among the following constitutes the maximum number of atoms?

- (a) 15×10^{25} molecules of SO_3
 (b) 2.5×10^{25} molecules of SO_2
 (c) 3×10^{25} molecules of O_2
 (d) 2.5×10^{25} molecules of N_2O_5

9. 300 mL of ethene is subjected to complete combustion. Calculate the volumes of O_2 required and CO_2 produced under same conditions.

- (a) 400 mL, 150 mL (b) 800 mL, 500 mL
 (c) 450 mL, 300 mL (d) 900 mL, 600 mL

Space for rough work

Space for rough work

10. The volume of gaseous products obtained by the thermal decomposition of 59.2 g of magnesium nitrate at STP is _____.

(a) 28 L (b) 22.4 L (c) 33.6 L (d) 54 L
11. The percentage composition of a compound is 80% C, 20% H. The weight of 1.5×10^{23} molecules of the compound is 7.5 g. Determine the molecular formula of the compound.

(a) C_3H_6 (b) C_2H_6 (c) C_2H_2 (d) C_2H_4
12. In which among the following reactions is the mole ratio of reactants 1:3?

(a) Reaction of calcium with water.
(b) Reduction of ferric oxide with coke.
(c) Synthesis of sulphur trioxide.
(d) Reaction between sodium hydroxide and sulphuric acid.
13. 175 g of calcium carbonate on thermal decomposition gives 33.6 L CO_2 at STP. Calculate the percentage purity of calcium carbonate.

(a) 80.7% (b) 85.7% (c) 90.7% (d) 95.7%
14. 96 g of 'X' contains 9×10^{23} molecules, and 0.32 g of gas 'Y' contains 1.2×10^{22} molecules. Compare the rate of diffusion (r) of X and Y at the same temperature and pressure.

(a) $r_Y = 3r_X$ (b) $r_Y = 2r_X$
(c) $r_X = 3r_Y$ (d) $r_X = 2r_Y$
15. 210 g of sodium bicarbonate is subjected to thermal decomposition in an open vessel and in a container fitted with a condenser separately. Calculate the change of loss in weight.

(a) 77.5 g (b) 22.5 g (c) 41 g (d) 55 g

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

- Assertion (A):** Two gram molecules of oxygen occupy 22.4 L at STP.
Reason (R): 1 mole of any gas contains Avogadro number of particles.

(a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
- Assertion (A):** Only Kelvin temperature is used in studying gas laws.
Reason (R): All values of temperature are positive in the Kelvin scale.

(a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
- Arrange the following compounds in the decreasing order of total number of atoms.

(A) Mercuric phosphate (B) Magnesium nitride
 (C) Cupric chlorite (D) Auric sulphite

(a) DACB (b) DABC (c) ADCB (d) ADBC
- On passing electricity, acidulated water gives oxygen and hydrogen. Oxygen formed, on treatment with carbon monoxide forms carbon dioxide, which on dissolution in water forms the corresponding acid. The acid formed on treatment with sodium hydroxide gives the corresponding salt and water. The aqueous solution of salt is then treated with potassium. Various types of reactions involved in the above process are given below. Arrange them in a proper sequence as given above.

(A) Double displacement (B) Displacement
 (C) Decomposition (D) Combination

(a) DCBA (b) CDAB
 (c) CDBA (d) DCAB
- Match the statements of Column A with those of Column B.

Column A (Relationship)	Column B (Gas Law)
(i) Pressure–volume	(A) Charles
(ii) Volume–temperature	(B) Boyle
(iii) Volume–number of molecules	(C) Graham
(iv) Rate of diffusion–density	(D) Avogadro

Space for rough work

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
(b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A)
(c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)
(d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
6. Which among the following compounds possesses trivalent cation and divalent negative radical?
(a) Calcium sulphide
(b) Aluminium nitrate
(c) Aluminium chromate
(d) Cupric hydroxide
7. The volume of a given mass of gas is 400 mL at 3 atm pressure. Calculate the new volume when pressure increases by 15% at constant temperature.
(a) 337 mL (b) 347 mL
(c) 327 mL (d) 357 mL
8. Which among the following possesses the least weight?
(a) 15×10^{22} molecules of O_2
(b) 0.15 moles of SO_3
(c) 2.24 L of SO_2 at STP
(d) 24×10^{22} atoms of He
9. 1 mole of a hydrocarbon requires maximum volume of oxygen for complete combustion under the same conditions. Identify the hydrocarbon.
(a) CH_4 (b) C_2H_6 (c) C_3H_6 (d) C_4H_6
10. Calculate the volume of CO_2 required for the complete combustion of 30 g of coke at STP, if the percentage conversion is 80%.
(a) 44.8 L (b) 50.4 L (c) 56 L (d) 61.6 L
11. The empirical formula of a compound is CH_4N . The empirical formula mass of the compound is equal to its vapour density. Which among the following compounds possesses the same number of carbon atoms as present in the above compound?
(a) Na_2CO_3 (b) $Ca(HCO_3)_2$
(c) $Al_2(CO_3)_3$ (d) $Fe(HCO_3)_3$
12. Which among the following is associated with same volume ratio of reactants as the formation of SO_3 from SO_2 and O_2 ?
(a) Synthesis of hydrogen chloride from H_2 and Cl_2 .
(b) Formation of CO_2 from CO and O_2 .
(c) Formation of NH_3 from its constituents.
(d) Synthesis of nitrogen monoxide from its constituents.

13. How many litres of CO_2 gas at STP are given by 70% pure 105 g of NaHCO_3 ?

- (a) 11.2 L (b) 18.2 L (c) 9.8 L (d) 14.8 L

14. Compare the rates of diffusion of oxygen, sulphur trioxide and ozone at constant temperature and pressure.

(a) $\frac{1}{2}:\frac{1}{5}:\frac{1}{3}$ (b) $\sqrt{2}:\sqrt{5}:\sqrt{3}$

(c) $\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{5}}:\frac{1}{\sqrt{3}}$ (d) 2:5:3

15. If 1 mole of each reactant is taken, then in which among the following reactants, on thermal decomposition, is the loss in weight found to be the maximum?

- (a) KNO_3 (b) KClO_3
(c) NaNO_3 (d) CaCO_3

Space for rough work

Answer Keys

Assessment Test I

1. (a) 2. (b) 3. (c) 4. (d) 5. (d) 6. (b) 7. (a) 8. (c) 9. (d) 10. (d)
11. (b) 12. (a) 13. (a) 14. (c) 15. (d)

Assessment Test II

1. (b) 2. (d) 3. (a) 4. (a) 5. (a) 6. (c) 7. (d) 8. (a) 9. (a) 10. (a)
11. (c) 12. (d) 13. (d) 14. (c) 15. (c)

Assessment Test III

1. (d) 2. (d) 3. (d) 4. (b) 5. (a) 6. (a) 7. (a) 8. (c) 9. (d) 10. (c)
11. (c) 12. (b) 13. (d) 14. (c) 15. (d)

Assessment Test IV

1. (d) 2. (c) 3. (a) 4. (b) 5. (c) 6. (c) 7. (d) 8. (b) 9. (a) 10. (c)
11. (b) 12. (d) 13. (a) 14. (a) 15. (b)

Water, Solutions, Solubility and Hydrogen; Metals and Non-metals

6



Reference: Coursebook - IIT Foundation Chemistry Class 9; Chapters - Water, Solutions, Solubility and Hydrogen; Metals and Non-metals; Page number - 7.1–7.14; 8.1–8.29

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. **Assertion (A):** Temporary hardness of water can be removed by boiling.

Reason (R): Bicarbonates of calcium and magnesium undergo thermal decomposition.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** Diamond is a good conductor of heat.

Reason (R): Carbon atoms are very closely packed in diamond.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Arrange the following saturated solutions in the increasing order of solubilities of the respective solutes.

(A) 125 g of solute is present in 625 g of solution 'A'.

(B) 20 g of solute is present in 180 g of solution 'B'.

(C) 30 g of solute is present in 210 g of solution 'C'.

(D) 15 g of solute is present in 120 g of solution 'D'.

- (a) ACBD (b) BDCA (c) CDBA (d) DCAB

4. Arrange the following general metallurgical processes in their proper order.

(A) Reduction of ore

(B) Concentration of ore

(C) Conversion of ore to oxide

(D) Dressing of ore

(E) Refining of ore

(a) ACDBE

(b) CBD AE

(c) DBCAE

(d) ECDBA

6.2 Chapter 6 Water, Solutions, Solubility and Hydrogen; Metals and Non-metals

5. Match the statements of Column A with those of Column B.

Column A (Substances)	Column B (Uses)
(i) Wood charcoal	(A) Black shoe polishes
(ii) Activated charcoal	(B) Extraction of metal
(iii) Lamp black	(C) Gas masks
(iv) Coke	(D) Removal of colour from sugar cane juice
(v) Animal charcoal	(E) Adsorbent material

(a) (i) → (C); (ii) → (E); (iii) → (A); (iv) → (B); (v) → (D)

(b) (i) → (C); (ii) → (A); (iii) → (E); (iv) → (B); (v) → (D)

(c) (i) → (C); (ii) → (A); (iii) → (B); (iv) → (E); (v) → (D)

(d) (i) → (B); (ii) → (E); (iii) → (C); (iv) → (A); (v) → (D)

6. Which of the following methods of concentration is inappropriate for an oxide ore?

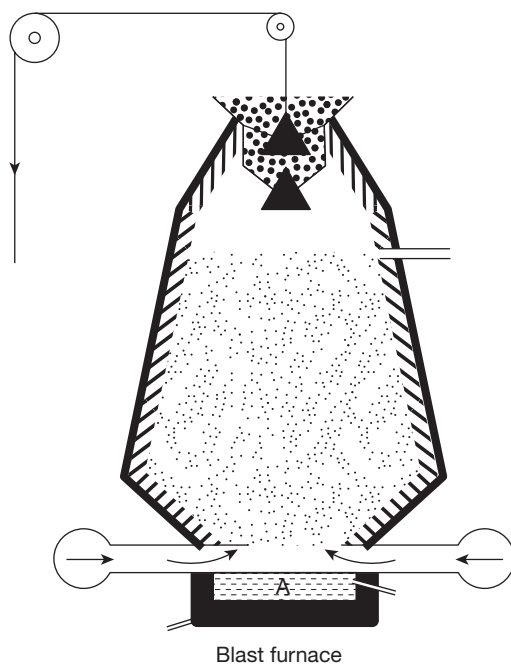
(a) Magnetic separation

(b) Gravity separation

(c) Froth floatation

(d) Chemical separation

7. Identify the substance present in region 'A' in the following figure.



(a) Molten slag

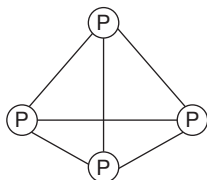
(b) Molten iron

(c) Coke

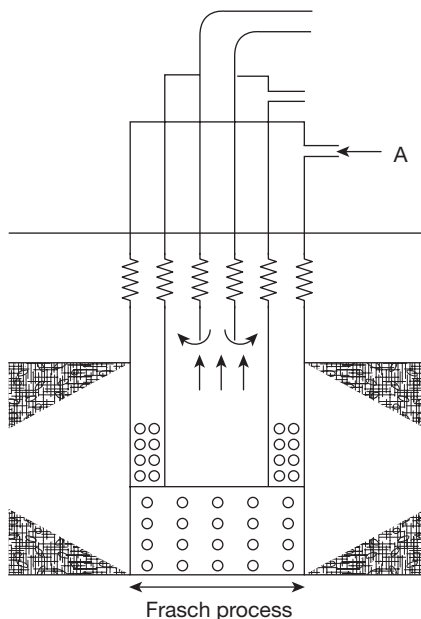
(d) Fe_2O_3

Space for rough work

8. A burning match stick, when introduced into a jar containing NH_3 and O_2 , burns with a pale blue flame. Identify one of the products of the reaction.
- (a) N_2 (b) N_2O (c) NO (d) NO_2
9. Identify the following properties that are not related to the given diagram.



- (a) Soft solid (b) Dark red
(c) Garlic smell (d) Poisonous
10. Identify 'A' from the figure given below.



- (a) Hot compressed air (b) Molten sulphur
(c) Super-heated steam (d) Solid roll of sulphur
11. The solubility of salt 'A' is 20 at a given temperature. How much of salt A is present in 600 g of saturated solution at that temperature?
(a) 100 g (b) 120 g (c) 20 g (d) 80 g
12. The molarity of NaOH solution is 5 M. Calculate the amount of solute required to be added to 1 L of the solution to saturate it at 25°C. The solubility of the solute is 20 at 25°C and the density of 5 M NaOH solution is 1.5 g/mL.
(a) 20 g (b) 50 g (c) 75 g (d) 60 g

Space for rough work

13. The atomic weight of a trivalent metal is x . Calculate its electrochemical equivalent.
- (a) $\frac{x}{96,500}$ (b) $\frac{x}{3 \times 96,500}$ (c) $\frac{x}{3}$ (d) $\frac{3x}{96,500}$
14. If ' a ' g of a divalent metal gets deposited at the cathode by passing 10 A of current for 30 min, calculate its atomic weight.
- (a) 214.4 a (b) 180 a
(c) 107.2 a (d) 53.6 a
15. A pale blue tinge in milk is due to _____.
(a) coagulation
(b) Tyndall effect
(c) Brownian movement of fat particles
(d) the ability of absorption of blue light by the fat particles

Space for rough work